ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Heating Effects

English Medium

EXERCISE

Single correct

1. Which of the following does not give metal oxide on heating

(A) $NaCO_3$ (B) K_2CO_3 (C) Rb_2CO_3 (D) All of these

HE0001

2. Which of the following metal bicarbonate will give metal oxide and CO_2 on heating

(A) $NaHCO_3$ (B) $Mg(HCO_3)_2$ (C) $KHCO_3$ (D) Rb_2CO_3

HE0002

3. Which of the following metal nitrate will give metal and oxygen on heating :

(A) KNO_3 (B) $NaNO_3$ (C) $AgNO_3$ (D) $RbNO_3$

HE0003

4. Which of the following nitrate will give N_2O on heating :

(A) NH_4NO_3 (B) NH_4NO_2 (C) $NaNO_3$ (D) $AgNO_3$

HE0004

5. Which of the following ammonium salt will not give acid on heating :

(A) $(NH_4)_2HPO_4$ (B) $(NH_4)_2MoO_4$ (C) $(NH_4)_2SO_4$ (D) NH_4Cl

HE0005

6. Which of the following halide will not give halogen gas on heating :

(A) $PbCl_4$ (B) $PbBr_4$ (C) Hg_2Cl_2 (D) All of these

HE0006

7. Select the correct statements

(A) Hydrated Co^{+2} salt is pink (B) Anhydrous Co^{+2} salt is of blue colour

(C) Hybridisation of $CoCl_2.6H_2O$ is sp^3d^2 (D) All of these

HE0007

8. Which of the following metal sulphate will give SO_2 and SO_3 both gaseous product on heating :

(A) $CuSO_4$ (B) $FeSO_4$ (C) $Fe_2(SO_4)_3$ (D) $CaSO_4$

HE0008

9. Which of the following compound is called dead burnt plaster :

(A) $CaSO_4.\frac{1}{2}H_2O$ (B) $CaSO_4.2H_2O$ (C) $CaSO_4$ (anhy.) (D) None of these

HE0009

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10. When NaH_2PO_4 is heated then which of the following compound is formed :

(A) $Na_4P_2O_7$ (B) Na_3PO_4 (C) HPO_3 (D) $NaPO_3$

HE0010

11. When $KMnO_4$ is heated then which of the following compound is formed :

(A) $K_2MnO_4 + MnO_2$ (B) $K_2MnO_4 + MnO$ (C) $MnO_2 + MnO$ (D) No change

HE0011

12. When CrO_3 is heated then ..... + ... are formed :

(A) Cr_2O_3, O_2 (B) CrO_2, O_2 (C) $Cr_2O_7^{-2}$, O_2 (D) None of these

HE0012

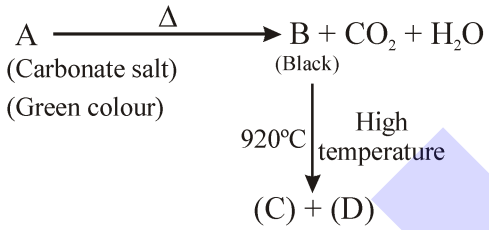
More than one may be correct

13. Which of the following metal carbonate will give of metal and oxyen on heating-

(A) Ag_2CO_3 (B) $HgCO_3$ (C) $(NH_4)_2CO_3$ (D) $PbCO_3$

HE0013

14.



Select the correct statements -

(A) Compound (A) is basic copper carbonate

(B) Compound (B) CuO

(C) Compound (C) is Cu_2O

(D) Compound (D) is paramagnetic in naturue

HE0014

15. When Ag_2CO_3 is heated then product will be -

(A) Ag_2O (B) Ag (C) O_2 (D) CO_2

HE0015

16. When compound A (orange red) is heated then green colour oxide of (B) is formed and inert gas (C) is formed then select the correct statements :

(A) Compound (A) is $(NH_4)_2\ Cr_2O_7$

(B) Compound (B) is used in fire works

(C) Gas C is N_2

(D) Heating effect of (A) is a type of intra molecular redox reaction

HE0016

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17. Which of the following hydrated salts will not become anhydrous on heating :

(A) $MgCl_2.6H_2O$ (B) $FeCl_3.6H_2O$ (C) $AlCl_3.6H_2O$ (D) $CoCl_2.6H_2O$

HE0017

18. Which of the following metal nitrate produce NO_2 on heating

(A) $Hg(NO_3)_2$ (B) $RbNO_3$ (C) $Pb(NO_3)_2$ (D) $Cu(NO_3)_2$

HE0018

19. Which of the following oxides turns yellow on heating and becomes white on cooling :

(A) ZnO (B) K_2O (C) PbO (D) Ag_2O

HE0019

Paragraph for Q. 20 & 21

$$\underset{\text{(Orange solid)}}{(A)} \xrightarrow{\Delta} \underset{\text{(yellow)}}{(B)} + \underset{\text{(green)}}{(C)} + \underset{\text{(Paramagnetic)}}{(D)}$$

20. Compound (A) is :

(A) $K_2Cr_2O_7$ (B) K_2CrO_4 (C) Cr_2O_3 (D) O_2

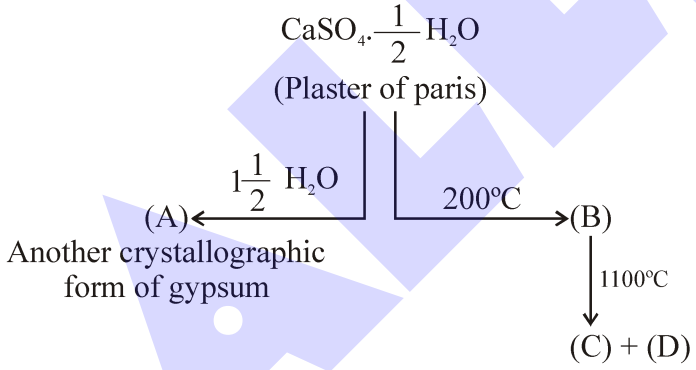
HE0020

21. Compound (C) is also obtained on heating of :

(A) $(NH_4)_2Cr_2O_7$ (B) NH_4ClO_4 (C) NH_4NO_3 (D) None of these

HE0020

Paragraph for Q. 22 & 23



22. Compound "A" is :

(A) $CaSO_4.2H_2O$ (B) $2CaSO_4.H_2O$ (C) $CaSO_4.3H_2O$ (D) $CaSO_4.5H_2O$

HE0021

23. Compound "C" and "D" are respectivelly :

(A) CaO + $CaSO_4$ (B) $CaSO_4$ + SO_2 (C) $CaSO_4$ + SO_3 (D) CaO + SO_3

HE0021

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Matrix match

24. Match the column

Column-I	Column-II
(A) $Be(NO_3)_2 \xrightarrow{\Delta}$	(P) Gives H_2O
(B) $HNO_2 \xrightarrow{\Delta}$	(Q) Oxyacid is obtained
(C) $NaH_2PO_4 \xrightarrow{\Delta}$	(R) Gives disproportionation reaction
(D) $H_3PO_3 \xrightarrow{\Delta}$	(S) Oxygen gas is evolved

HE0022

25.

Column-I (Compound)	Column-II (Products on heating)
(A) NH_4ClO_4	(P) CO_2 gas is evolved
(B) $Mg(OAc)_2$	(Q) H_2 gas is evolved
(C) HCOONa (above 350ºC)	(R) N_2 gas is evolved
(D) $(NH_4)_2CO_3$	(S) Same gas is evolved which is obtained by heating $(NH_4)_2SO_4$
	(T) Intra molecular redox reaction

HE0023

Integer

26. When calamine is heated then a product (A) is formed then find the total number of following options are correct for compound (A) -

(i) Compound (A) is white in cold conditions

(ii) Compound (A) is yellow in hot conditions

(iii) Compound (A) is called phillosopher's wool

(iv) Compound (A) when combined with CoO, then compound (B) is formed & colour of new compound (B) is green

(v) Compound (B) is called Rinmann's green

HE0024

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27. $NaNO_3 \xrightarrow{500°C}$ (A) + (B)

(A) $\xrightarrow{800°C}$ (C) + (D) + (E)

Find the number of correct statements

(1) Compound (B) is paramagnetic in nature

(2) Compound (B) when undergoes dimerisation then dimer product is diamagnetic in nature

(3) Bond order of compound (B) is two

(4) D is N_2 gas

(5) Compound B and E are same gas

HE0025

28. (A) $\xrightarrow{\Delta}$ (B) + (C) + (D)
(dark purple almost black) (green) (black)

Find the number of correct statements

(1) Compound B is K_2MnO_4

(2) Compound C is MnO_2

(3) Compound D is O_2

(4) Compound B is paramagnetic in nature

(5) Compound D has two unpaired electron in bonding molecular orbital

HE0026

29. Total number of compounds undergoes dispropornation redox reaction on heating
MnO_2, HOCl, H_3PO_3, HNO_2, CrO_5, $HClO_3$

HE0027

30. On strong heating of H_3PO_4 and H_3BO_3, sum of oxidation number of P & B in the final product obtained is

HE0028

ANSWER-KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	C	A	B	C	D	B	C	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A, B	A,B,C,D	B,C,D	A,B,C,D	A, B, C	A,C,D	A, B	A
Que.	21	22	23	24						
Ans.	A	A	D	(A)→(S);(B)→(P,Q,R);(C)→(P);(D)→(Q,R)						
Que.	25					26	27	28	29	30
Ans.	(A)→(R,T);(B)→(P);(C)→(P,Q,T);(D)→(P,S)					5	4	4	4	8

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